ECDLP lead atlas: EcGFp5 quotient-solve update, 2026-11-01

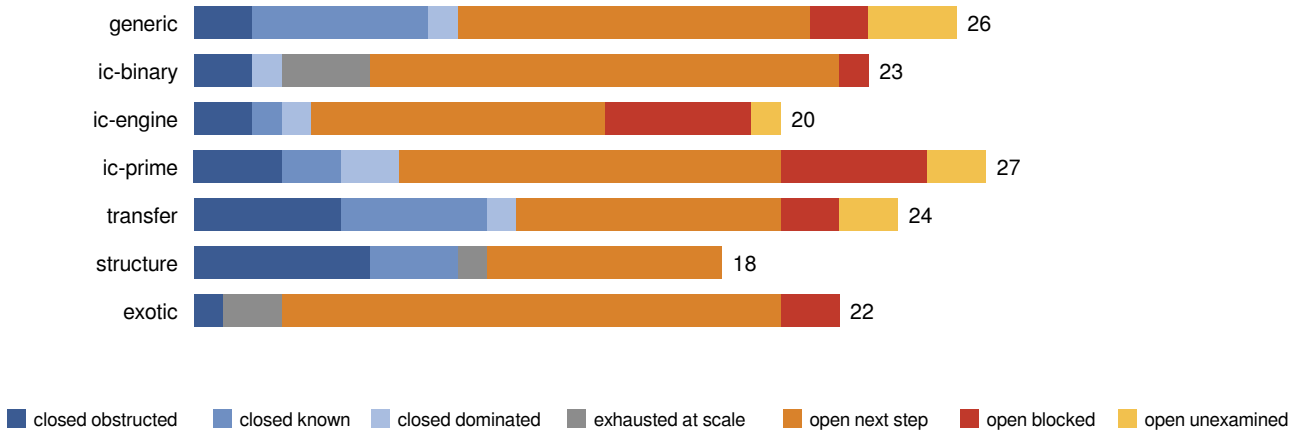
Question. The crypto quotient-solve experiment is now complete. What does it decide for `LEAD-ICP-gfp5-symmetrized-gaudry`?

Scope. This is a single-lead fold, not a full atlas review. `--delta` was not run and `REVIEW_STATE.json` is not stamped; the next full review picks up the other changes since the 2026-10-30 pins. The fold cites one experiment:

- `crypto:research/ecgfp5_quotient_solve_20261101/` (crypto PR #3267). Its protocol was frozen before the first run and amended five times (A1–A5, each recorded in `PROTOCOL.md`).
- That experiment was designed in `crypto:research/ecgfp5_lead_evaluation_20261031/`.

Status of each statement. The disposition is derived from the cited evidence. Numbers are measured values copied from that report. The fold changes no hypothesis, experiment or goal status.

ECDLP lead atlas, 2026-11-01: 160 leads by family and disposition



Result

- `LEAD-ICP-gfp5-symmetrized-gaudry`**: open_next_step → closed_dominated** (obstruction kind `measured`).
  - The F_p-rational (Z/2)^4 x| S_5 quotient of KN-OPEN-9b4a2b was solved completely at m = 5 with msolve 0.6.5. The three toy primes were p' = 4111, 262231 and 33554371.
  - D = 2^16 in every cell, which confirms D = 2^{(m−1)^2}. The planted point was recovered every time. Peak RSS was 4.7–6.5 GiB.
  - Whole-solve per-system cost was c_cyc = 2^46.30, 2^46.56 and 2^46.63. That meets the frozen G1 CLOSE threshold (≥ 2^46) at every prime.
  - The margin is small: 0.30–0.63 bits. F4 alone cost 2^44.7–2^45.3, which would sit in G1's not_measured band. About two-thirds of c is msolve's rational parametrisation.
  - Five solver probes at p' = 4111 each failed the improvement bar of 6,900 CPU-s (W0 F4 / 1.5):
    - weighted F4 (2,2,2,2,2,1) finished at 1.16× W0 CPU, with D = 2^21;
    - weighted (1,1,1,1,1,2) and (1,2,3,4,5,5) hit the wall cap;
    - Singular sba F5a hit its cap;
    - Singular sba F5b was stopped by the owner past the bar.
- **Why closed_dominated and not closed_obstructed.** A finite toy-prime cost measurement is not a disproof. The lead stays in the pursuit queue (README, "Pursuit invariant"). It reopens on any of these:
  - a finish cheaper than the rational parametrisation (FGLM to a lex eliminant, or the eliminant only) brings c_cyc to ≤ 2^38 on ≥ 3 primes;
  - a matrix-F5/GVW-style F4 lowers F4 CPU below W0 / 1.5;

  - a construction gives D < 2^16 at m = 5.
  - Separately, H-GFPN-6cc6aa (degree-4 Macaulay rank) is still untested.
- **Tallies after the fold.** 160 leads:
  - closed: 21 obstructed, 17 known, 6 dominated, 6 exhausted at scale;
  - open: 85 next step, 17 blocked, 8 unexamined.

What this does not say

- It does not say the EcGFp5 symmetrized decomposition is impossible.
- It does not compare the quotient solve with rho end to end. c is a per-system stage cost.
- The larger primes on EKS were not run.

Files

- `dispositions.svg` and `dispositions.json`: drawn by `figures.py` from the committed rows.
- `report.pdf`: rendered from this README by `build_pdf.py`.